Milestone 2 | CIRES

Citizen Issue Reporting and Escalation System

From a resident's report to a recorded resolution

Requirements engineering and system analysis

One area, two issue categories

The pilot serves residents, assigned officers and a supervisor.

In the pilot

Road and drainage issues

Report submission, assignment, tracking and resolution

One escalation per overdue case

Outside the pilot

Emergency dispatch and other issue categories

District-wide rollout or agency integrations

SMS alerts and multi-level escalation

Five places the reporting trail can break

These design concerns came from Milestone 1 scenario analysis.

PP1 Receipt	The resident may leave without a reference for follow-up.
PP2 Ownership	A case may have no clearly assigned officer or deadline.
PP3 Visibility	The resident may need to call again to learn what happened.
PP4 Delay	The supervisor may learn about a delay only after a complaint.
PP5 Closure	A resolved status may say little about the action taken.

Reporting, assignment and progress

Each requirement links to a Milestone 1 design concern.

FR-01 • Must / PP1	The system shall accept a report with category, location and description and return a unique reference.
FR-02 • Must / PP2	The system shall let the supervisor assign a Received report to a pilot officer with a future deadline.
FR-03 • Must / PP3	The system shall show the owner the current status and dated public updates for their report.
FR-04 • Must / PP2	The system shall let the assigned officer change an Assigned report to In progress.

Progress, escalation and review

Each requirement links to a Milestone 1 design concern.

FR-05 • Must / PP4	The system shall flag each overdue Assigned or In progress report once and show it in the supervisor’s queue.
FR-06 • Must / PP5	The system shall require a public action summary before the assigned officer resolves an In progress report.
FR-07 • Should / PP2	The system shall filter the supervisor’s report list by category and status.
FR-08 • Could / PP5	The system shall export weekly counts of resolved reports by category without resident identifiers.

Access controls and response time

Proposed acceptance targets for the pilot.

NFR-01 Access	All core actions require sign-in. Cross-account reads and unauthorised updates return no case data and make no changes.
NFR-02 Privacy	Resident views and exports expose zero private contact fields or internal staff notes.
NFR-03 Speed	At least 95% of report detail requests finish within 2 seconds in 50 trials with 1,000 stored cases.

Access for residents and pilot uptime

Proposed acceptance targets for the pilot.

NFR-04 Accessibility	Core forms support keyboard operation, visible labels and 4.5:1 contrast for normal text.
NFR-05 Usability	Two first-time resident testers can each submit a valid report within 2 minutes without assistance.
NFR-06 Availability	At least 99% of scheduled five-minute checks succeed during a seven-day pilot observation window.

Submission and assignment

User stories with MoSCoW priority, effort estimate and acceptance criteria.

US-01 / Must / 3 pts

As a resident, I want to submit a road or drainage report so I can receive proof that it was received.

Given a signed-in resident enters valid details
When they submit
Then one Received report is saved and its reference appears.

PP1 / FR-01

US-02 / Must / 3 pts

As a supervisor, I want to assign a report with a due date so I can make ownership clear.

Given an unassigned Received report exists
When the supervisor selects an officer and future deadline
Then the officer sees it in their queue and the report becomes Assigned.

PP2 / FR-02

Tracking and officer updates

User stories with MoSCoW priority, effort estimate and acceptance criteria.

US-03 / Must / 2 pts

As a resident, I want to check my report so I can see its progress without calling.

Given a resident owns an assigned report
When they open its reference
Then they see its current status and public history.

PP3 / FR-03

US-04 / Must / 2 pts

As a assigned officer, I want to start work on my case so I can show that action is underway.

Given a report is assigned to the officer
When they mark it In progress
Then the new status and timestamp appear in its history.

PP2 / FR-04

Escalation and resolution

User stories with MoSCoW priority, effort estimate and acceptance criteria.

US-05 / Must / 3 pts

As a supervisor, I want to see overdue cases in one queue so I can follow up before they are forgotten.

Given an open assigned report has passed its deadline
When the overdue check runs twice
Then exactly one escalation is recorded and visible to the supervisor.

PP4 / FR-05

US-06 / Must / 3 pts

As a assigned officer, I want to close a case with an action summary so I can explain the outcome to the resident.

Given an In progress report belongs to the officer
When they save a nonblank action summary and resolve it
Then the report becomes Resolved and the resident can read the summary.

PP5 / FR-06

Queue filters and reporting

User stories with MoSCoW priority, effort estimate and acceptance criteria.

US-07 / Should / 5 pts

As a supervisor, I want to filter the officer queue by category and status so I can find cases for follow-up.

Given the queue contains mixed categories and statuses
When the supervisor applies both filters
Then only matching reports are shown.

PP2 / FR-07

US-08 / Could / 5 pts

As a supervisor, I want to export a weekly closure summary so I can review the pilot's workload.

Given resolved cases exist within and outside the selected week
When the supervisor exports that week
Then the CSV contains only counts by category for that week.

PP5 / FR-08

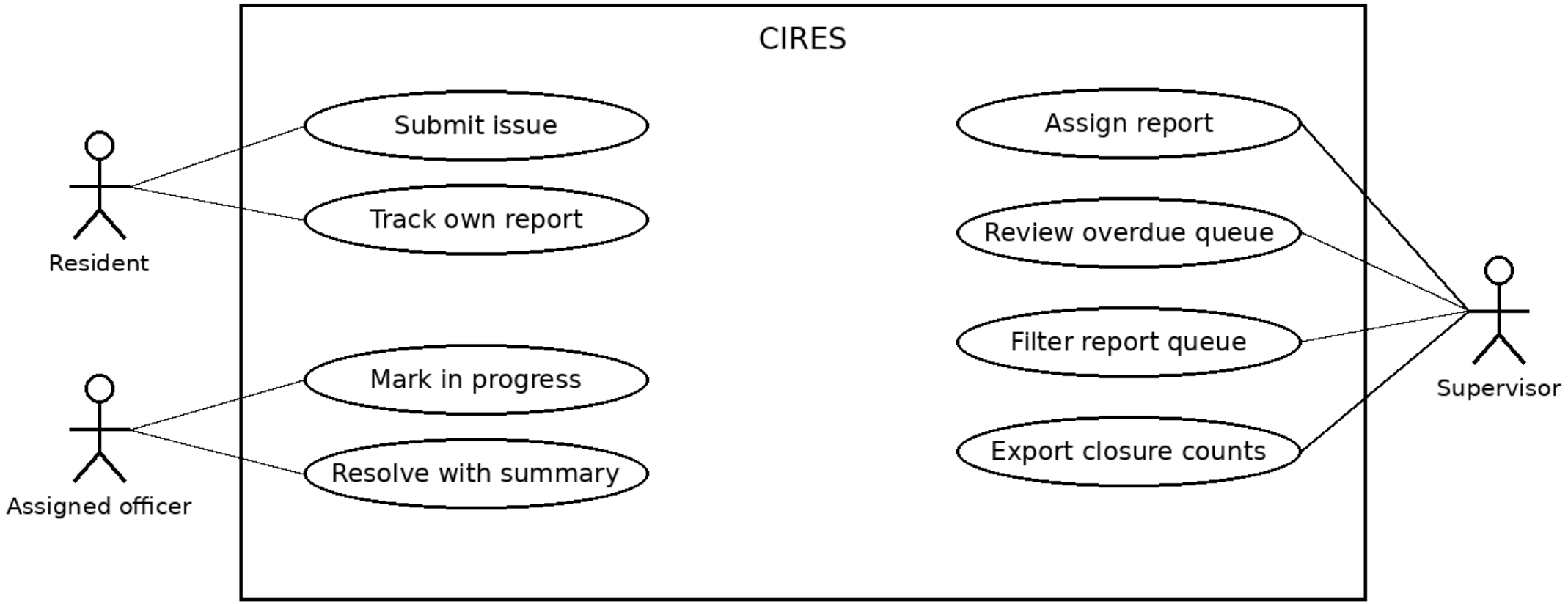
The core backlog spans two sessions

Must-have work totals 16 of 26 estimated points, about 62%.

Must / 16 pts	Session 4: submit, assign and track (8 pts). Session 5: update, escalate and resolve (8 pts).
Should / 5 pts	US-07. Filter the supervisor’s queue by category and status.
Could / 5 pts	US-08. Export a weekly closure count by category.
Won’t / this release	Emergency dispatch, SMS and multi-level escalation.

CIRES use-case diagram

Actors sit outside the system boundary. Solid lines show who participates.



UC-01 Submit an issue

Primary actor / Resident • FR-01

Precondition	The resident is signed in and the location is in the pilot area.
Main flow	The resident selects Road or Drainage, enters the location and description, and submits.
Alternative flow	Missing fields or an unsupported location produce field errors. No report is saved.
Postcondition	CIRES saves a Received report and displays a unique reference.

UC-02 Review an overdue case

Primary actor / Supervisor • FR-02, FR-05 and FR-06

Precondition	The report has an assigned officer and a due timestamp.
Main flow	A scheduled check flags the overdue open report. The supervisor opens it from the overdue queue and reviews its history.
Alternative flow	Resolved cases are skipped. A repeated check retains the original escalation and timestamp.
Postcondition	The overdue case stays visible until resolution. Its escalation remains in the history.

How a report moves through CIRES

Escalation flags an overdue report while its work status stays open.

Received → **Assigned** → **In progress** → **Resolved**

Overdue flag

Assigned or In progress + deadline passed. One escalation per report.

Every requirement has a test

Story acceptance criteria define the observable result.

PP1 Receipt	FR-01 → US-01 → T01: valid report returns a unique reference.
PP2 Ownership	FR-02 / 04 / 07 → US-02 / 04 / 07 → T02 assignment, T04 progress, T07 filters.
PP3 Visibility	FR-03 → US-03 → T03: owner sees the status and public history.
PP4 Delay	FR-05 → US-05 → T05: repeated checks produce one escalation.
PP5 Closure	FR-06 / 08 → US-06 / 08 → T06 action summary, T08 weekly counts.

Two user reviews for the Must list

Validation status / User feedback is pending.

Resident review

Walk through reporting a blocked drain and finding its status.

Which update would tell you whether anyone had acted?

Assigned officer review

Walk through an assigned road issue approaching its deadline.

Which details would you need to decide what to do next?

Milestone 2 deliverables

Course basis / Session 2, slides 27–36 and Student Project Guide §6.

Requirements package	SRS PDF, numbered requirements, traceability and eight stories with acceptance criteria.
Draw.io	Editable UML source and matching monochrome PNG.
Jira or Trello	The CSV contains the backlog for import. The live board link and screenshot are pending.
User review	Two real user reviews and a record of requested changes are pending.
GitHub release	Store the package in docs/m2-cires/ and create the m2 release after completing the evidence.